MTH 335(A) Dr. James Ashe

Final Exam Thursday 2^{nd} May, 2013 **Name:**

1. Let $A = \{1, 2, 3, 5\}$ and $B = \{a, b, 2, c, 5\}$

a. Find $A \cap B$.

b. Find $A \cup B$.

c. Find $B - A$.

d. Find $A - B$.

2. Let A be a nonempty subset of B. Which of the following is true?

(a) $A \cap B = B$.

(b) $A \cup B = A$.

(c) $A \cup B = B$.

(d) $A \cap B = \emptyset$.

3. Let $A = \{3, 5\}$ and $B = \{a, b, 5\}$.

a. List all possible subsets of B.

b. Find $A \times B$.

4. Prove that $A \cup \{\emptyset\} = A$.

5. Consider the map $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ where $f(x, y) = xy$. Prove that f is onto.

6. Define the relation $\mathcal{R}$ on the real number set $\mathbb{R}$ as $x\mathcal{R}y \iff xy \geq 0$.

a. Show that $\mathcal{R}$ is reflexive and symmetric.

b. Is $\mathcal{R}$ an equivalence relation? Explain.

7. Let X be a nonempty set. Define the following relation on the set of subsets of X as follows $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$.

a. Show that $\mathcal{R}$ is symmetric.

b. Show that $\mathcal{R}$ is not reflexive.

8. Let $f : A \to B$ and $g : B \to C$ be two functions.

a. If $g \circ f$ is one-to-one then

(a) f is onto.
(b) f is one-to-one.
(c) g is one-to-one.
(d) Both f and g are one-to-one.
(e) None of the above.

b. If $|A| > |B|$ and $A \neq \emptyset$ can f be one-to-one?

9. Let S_3 be the symmetric group on three elements.

a. Which of the following is true?

(a) S_3 has at least one non-abelian subgroup.
(b) All proper subgroups of S_3 are abelian.
(c) S_3 is abelian.
(d) S_3 is cyclic.

b. What is the order of S_3 and S_4?

10. Let $f : G_1 \to G_2$ be a group homomorphism and let $H \leq G_2$. Recall that $f^{-1}(H) = \{g \in G_1 | f(g) \in H\}$.

a. Which of the following is true?

(a) $f^{-1}(H)$ consists only of the identity of G_1.
(b) $f^{-1}(H)$ is a subgroup of G_1.
(c) $gf^{-1}(H) = f^{-1}(H)g$ for all $g \in G_1$ (the left and right cosets are equal).
(d) $f^{-1}(H)$ is not a subgroup of G_1

b. Let $a, b \in G_1$. Does $f(ab) = f(a)f(b)$?

11. Let G be a finite group of order 12 and $g \in G$.

a. The order of g cannot be:

(a) 8
(b) 6
(c) 4
(d) 3

b. Let $|H| = 6$. Does Lagrange's theorem imply that $H \leq G$?

12. Let G be a group of order 4. Which of the following is true?

(a) G is a cyclic group.
(b) G is $\mathbb{Z}_2 \times \mathbb{Z}_2$.
(c) G is $\mathbb{Z}_4$.
(d) G is abelian.

13. Let $G = \{e, a_1, a_2, a_3, a_4, a_5\}$ be a group where e is the identity.

a. Find a subgroup of G.

b. Find a subgroup of G distinct from the group given for part **(a)**.

c. Can $|a_2| = 5$?

14. Let H and K be subgroups be non-trivial subgroups of G. Prove or disprove that $H \cap K$ is a subgroup of G.

15. Let R be a ring and $a, b \in R$.

a. Which statement is true?

(a) If $ab = 0$ then $a = 0$ or $b = 0$.
(b) If $a^2 = 0$ then $a = 0$.
(c) If $a \neq 0$ and $b \neq 0$ then $ab \neq 0$.
(d) None of the above.

b. Show that $\mathbb{Q}$, the set of rational numbers, is a ring under addition and multiplication.